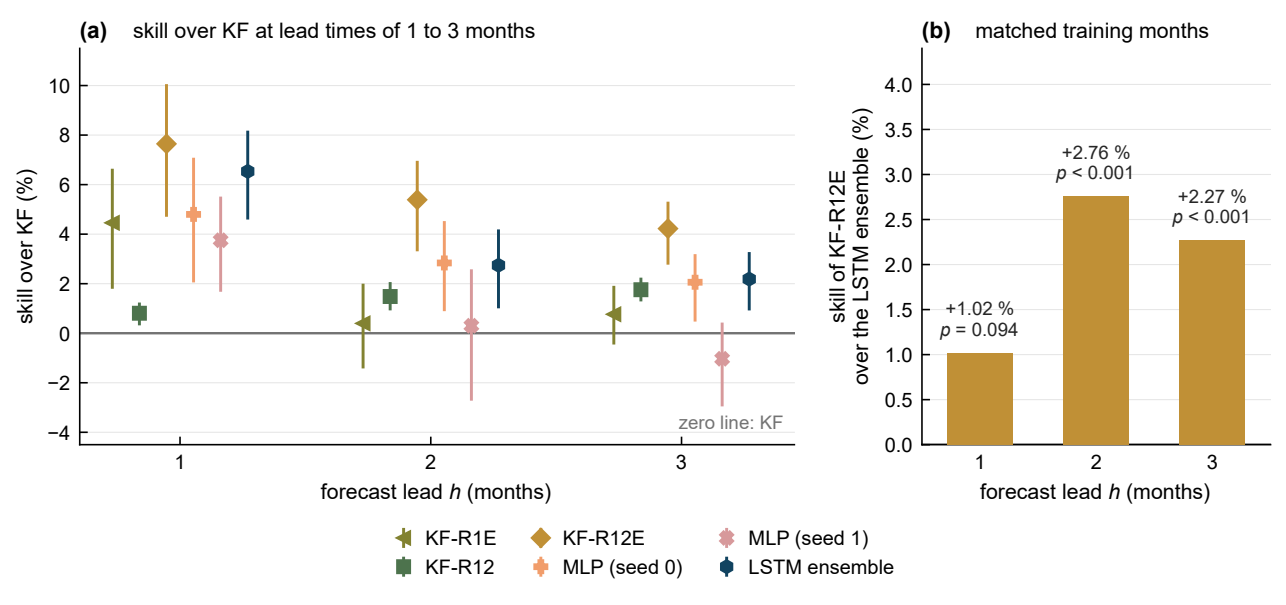

(a)
skill over KF at lead times of 1 to 3 months
skill over KF (%)
10
8
6
4
2
0
−2
−4
zero line: KF
1
2
3
forecast lead h (months)
(b)
matched training months
skill of KF-R12E
over the LSTM ensemble (%)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
+1.02 %
p = 0.094
+2.76 %
p < 0.001
+2.27 %
p < 0.001
1
2
3
forecast lead h (months)
KF-R1E
KF-R12
KF-R12E
MLP (seed 0)
MLP (seed 1)
LSTM ensemble